

DHULAKSHAN KANNAN

Data Science Undergraduate

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PROFESSIONAL SUMMARY

Data Science undergraduate with a solid foundation in statistics, time series analysis, and machine learning, plus academic exposure to data engineering, financial analytics, and deep learning. Hands-on experience with R, Python, SQL, and Power BI through forecasting, network analytics, and cloud data projects, along with exposure to deep learning frameworks such as TensorFlow, Keras and PyTorch. Aspiring AI researcher, driven to build data-driven solutions with real-world impact. Practical experience extends to building end-to-end AI systems with YOLOv8 and MLOps pipelines using MLflow, DVC, Docker, and GitHub Actions CI/CD, along with data engineering work involving Apache Kafka, Airflow, and PySpark.

EDUCATION

BSc (Hons) in Data Science **2024 - 2027**

Coventry University | National Institute of Business Management (NIBM)

Secondary Education — GCE Advanced Level (Physical Science Stream) **2020 - 2022 (2023)**

D.S. Senanayake College, Colombo 07

Primary Education — GCE Ordinary Level **2014 - 2019**

K/Hindu National College, Pussellawa

TECHNICAL SKILLS

- **Machine Learning and AI:** Supervised Learning, Unsupervised Learning, Reinforcement Learning, Deep Learning, Time Series Analysis, PyTorch, TensorFlow, YOLOv8
- **Data Engineering and Big Data:** Data Pipelines, Data Processing, ETL Concepts, Big Data Analytics, Apache Kafka, Apache Airflow
- **Programming and Tools:** Python (Pandas, NumPy, Scikit-learn), R, SQL & NoSQL, Cloud Computing, Docker, MLflow
- **Data Visualization:** Power BI, Matplotlib, Seaborn
- **Mathematics and Analytics:** Statistics, Probability, Calculus, Linear Algebra

RESEARCH

Daily-Based PM2.5 Pollution Pattern Identification using Fuzzy C-Means Clustering: A Case Study of the Colombo Municipal Council Area

- Built a probabilistic preprocessing and clustering framework using Normal, Lognormal, Gamma, and Weibull distribution fitting, AIC-based model selection, Min-Max normalization, and Fuzzy C-Means clustering on three years of PM2.5 data collected from 12 monitoring stations within the Colombo Municipal Council area.
- Engineered an end-to-end pipeline in Python, transforming historical air quality observations into representative expected-value datasets for robust clustering analysis.
- Applied FCSETS-based stability validation to determine the optimal clustering configuration, generating fuzzy memberships, cluster centers, and pollution pattern labels for comprehensive interpretation.
- Identified five distinct daily PM2.5 pollution patterns, revealing temporal trends, weekday-weekend variations, and station-level behaviour to support data-driven urban air quality management.

ACADEMIC PROJECTS

CrowdSafe AI — Real-Time Crowd Monitoring & Stampede Detection | Deep Learning & MLOps

- Developed an end-to-end crowd safety system: fine-tuned a YOLOv8 Nano model on a custom Roboflow crowd dataset via Google Colab (50 epochs, T4 GPU) for real-time people detection.
- Built a Streamlit dashboard integrating Gaussian crowd density heatmaps, Farneback Dense Optical Flow for movement analysis, and a 4×3 grid-based zone risk scoring system for multi-signal stampede detection.
- Engineered a full MLOps pipeline with MLflow for experiment tracking, DVC for data/model versioning, and GitHub Actions for CI/CD automation.
- Containerized the application with Docker and exposed a Flask REST API for model serving, with drift detection to monitor model performance over time.
- Resolved multiple environment-level engineering challenges, including PowerShell/Windows path handling, MLflow tracking store issues, and CI/CD pipeline configuration.

Tech Stack: Python, YOLOv8, Streamlit, MLflow, DVC, Docker, Flask, GitHub Actions, OpenCV

CityPulse — Weather Analytics Pipeline | Data Engineering

- Built an end-to-end data engineering pipeline for weather analytics, integrating Apache Kafka for stream ingestion, PostgreSQL for storage, and Apache Airflow for workflow orchestration.
- Designed and implemented PySpark transformation jobs to process and structure incoming weather data for downstream analysis.
- Containerized the full multi-service stack (Kafka, PostgreSQL, Airflow, PySpark) with Docker Compose for reproducible, single-command local deployment.
- Resolved a critical infrastructure issue by migrating from a paywalled Kafka image to the open-source apache/kafka image, reconfiguring KRaft environment variables to maintain pipeline stability.

Tech Stack: Docker, Apache Airflow, Apache Kafka, PySpark, PostgreSQL

SUB PROJECTS

- **Airbnb NYC ELT Data Pipeline:** Normalized ELT pipeline for NYC Airbnb listings data, using a PostgreSQL star schema and Apache Airflow for automated orchestration.
- **IoT-Based Temperature & Humidity Monitoring Dashboard:** Real-time IoT monitoring dashboard built with Python Dash, featuring live visualization and alerts.
- **USD/LKR Exchange Rate Forecasting (SARIMA):** Forecasted monthly USD/LKR exchange rates using a SARIMA model on 10 years of data, with stationarity testing and residual diagnostics.
- **Geospatial Population Visualization Dashboard (Bokeh):** Interactive choropleth dashboard visualizing world population trends, with map tiles, a year slider, and linked trend charts.

WORK EXPERIENCE

Sales & Operations Manager | Ceylon Hampers Pvt Ltd

May 2024 – June 2025

- Led end-to-end sales and operational activities for a gifting business, managing order processing, inventory coordination, and delivery timelines.
- Analyzed sales performance and operational data to identify trends and support data-driven business decisions.
- Coordinated with suppliers and internal teams to optimize operational efficiency and maintain service quality.
- Built and maintained customer relationships, resolving inquiries and driving repeat orders through improved engagement strategies.